PHYS 6031
Spring 2024
Homework 5
Due: Tuesday 27 February 2024 in class

1. Determine all electrostatic multipole moments q_{lm} for the following charge distribution: A line charge λ per unit length uniformly distributed around a ring of radius a (the total charge is $Q = 2\pi a\lambda$). The ring is centered on the z axis and lies in the $x - y$ plane.

2. Determine the electrostatic multipole moments q_{lm} for the following charge distribution: A line charge λ per unit length uniformly distributed along the x axis from $x = -a/2$ to $x = a/2$, where a is a constant.

3. A sphere of radius R is made of a material with dielectric constant ϵ/ϵ_0. It is placed in a constant external electric field of strength $\vec{\boldsymbol{E_0}}$. Determine the electrostatic potential everywhere.

4. A point dipole moment $\vec{\boldsymbol{p_0}}$ is placed at the center of a dielectric sphere of radius a and dielectric constant ϵ/ϵ_0. Find the electrostatic potential everywhere.